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This report should be shared with the entire technical team and leadership, not only to learn from the incident but also to update the incident response procedure for future occurrences. Businesses that maintain the habit of periodic reviews, timely vulnerability patching, and security awareness training for employees tend to significantly reduce the number of severe incidents over the long term. No system is ever completely secure, but a clear incident response process that is regularly rehearsed helps minimize downtime and damage when an incident actually occurs. In addition to incident response, patch management is another critical pillar of an organization's information security program.

Whenever a software vendor releases a new patch, the operations team needs to assess the severity of the vulnerability based on its CVSS score, scope of impact, and real-world exploitability before scheduling deployment. Vulnerabilities with a score of nine or higher that already have public exploit code must be patched within seventy-two hours, while medium-severity vulnerabilities can be scheduled into the monthly maintenance cycle. Before deploying a patch to the production environment, the technical team should test it in a staging environment with a similar configuration to catch compatibility issues early. A patch applied hastily without testing can cause more severe service disruption than the vulnerability it was meant to fix.

Every change must be recorded in the change management system, including the version before and after patching, who performed it, the deployment time, and the rollback plan in case the patch causes errors. For systems that cannot be patched immediately due to operational constraints, the security team must apply temporary mitigations such as network isolation, restricting access, or enabling intrusion detection rules specific to that vulnerability, while closely monitoring the system until the official patch can be applied. The IT asset inventory needs to be kept up to date so the operations team always knows exactly which software version is running where, avoiding the risk of missing servers during vulnerability reviews. A periodic patching status report should be presented to leadership every quarter, including the number of remaining critical vulnerabilities, the average time to patch, and the trend compared to the previous period.

An effective patch management program not only helps reduce the attack surface but also serves as important evidence when a business needs to demonstrate compliance with information security standards to partners or regulators. Adequate investment in vulnerability scanning and patch deployment automation will save the technical team significant time compared to handling each server manually one by one. When a small or medium-sized business detects unusual activity on its network, the incident response team must follow a five-step process: identification, containment, eradication, recovery, and lessons learned. During identification, administrators collect logs from the firewall, web server, and intrusion detection system to confirm whether this is a real incident or merely a false alarm.

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All initial evidence, including timestamps, source IP addresses, and related accounts, must be preserved intact before making any changes to the affected system. In a typical case, the operations team detects unusual traffic from the address 198.51.100.23 attempting to exploit vulnerability CVE-2024-21412 on a server running software version 3.4.1. The attacker repeatedly sends requests to the path https://portal.noi-bo.example.com/api/v1/upload in an attempt to upload an unauthorized executable file. As soon as this is detected, the response team must isolate the affected server from the internal network, block the source IP address at the edge firewall, and disable the service account being abused, while making sure not to delete any log records, since they serve as critical evidence for later investigation.

The eradication step requires identifying the root cause. If the cause is an unpatched vulnerability, the technical team must apply the official patch from the vendor, review access permission configurations, and inspect all related servers to make sure no backdoor has been planted. Rescanning the entire system with an up-to-date malware detection tool is mandatory before moving to the recovery step. During recovery, the system can only be brought back online after confirming there are no remaining signs of intrusion, administrative account passwords have been changed, and clean backups are ready to restore data if needed.

The operations team should closely monitor the system for at least seventy-two hours after recovery to detect any early sign of reinfection. Finally, the lessons-learned step is where the entire incident is recorded into a detailed report: the time of detection, the time of containment, the root cause, the estimated damage, and the preventive measures to be implemented going forward. This report should be shared with the entire technical team and leadership, not only to learn from the incident but also to update the incident response procedure for future occurrences. Businesses that maintain the habit of periodic reviews, timely vulnerability patching, and security awareness training for employees tend to significantly reduce the number of severe incidents over the long term.

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